Ragas evaluation: outcome and scores

Run 20261111_205750 | 30 documents | 1,136 chunks | 10 source questions + 3 status checks | final top 5

10 / 10	8 / 10	8
Expected document found by each search mode	Source questions answered by chat on first attempt	Answers eligible for Ragas judging

The end-to-end chat returned HTTP 502 on Q08 and Q09, including one retry each. Their missing Ragas values are failures to answer, not scores of zero. The three separate search-mode calls succeeded for both questions.

Ragas scores for the eight answered cases

Metric	Mean	Median	What it checks
Faithfulness	0.927	1.000	Are answer claims supported by selected chunks?
Answer Correctness	0.650	0.528	Does the answer agree with the reviewed reference?
Context Precision	0.934	1.000	Are useful selected chunks near the top?
Context Recall	1.000	1.000	Do selected chunks cover the reference claims?
Answer Relevancy	0.921	0.931	Does the answer address the question?

Reading the scores

Faithfulness 0.927 and context precision 0.934 suggest the eight completed answers usually used relevant evidence. Context recall is 1.000, but the references mostly contain one main claim, so this small set gives recall limited power to distinguish cases. Answer relevancy 0.921 indicates generally focused answers; it does not establish factual correctness.

Answer correctness averages 0.650. This is a judge score, not a statement that 65% of answers are correct. Several low scores conflict with a direct reading of the cited passage, so inspect individual answers before using the mean as a quality target.

Ragas judged the selected chat evidence saved in LangGraph checkpoints. These scores were not computed separately for lexical, semantic, and hybrid /search calls.

Question-level answer evaluation

Each row is the first /chat attempt. A dash means the chat request failed before Ragas could score an answer.

ID	Status	Faith.	Correct.	Precision	Recall	Relev.
Q01	answered	1.000	0.964	1.000	1.000	1.000
Q02	answered	1.000	0.512	0.806	1.000	0.918
Q03	answered	1.000	0.428	0.750	1.000	0.882
Q04	answered	1.000	0.951	1.000	1.000	0.996
Q05	answered	1.000	0.988	1.000	1.000	1.000
Q06	answered	0.667	0.390	1.000	1.000	0.944
Q07	answered	0.750	0.544	0.917	1.000	0.708
Q08	error	-	-	-	-	-
Q09	error	-	-	-	-	-
Q10	answered	1.000	0.425	1.000	1.000	0.916

Manual review of low scores

Q06 (correctness 0.390; faithfulness 0.667): the answer names Schedule (5) Items (3), (4), and (5), matching the reference and cited passage. The judge scores appear too low for that direct match.

Q10 (correctness 0.425): the answer repeats the cited taxicab definition with introductory wording. This is another apparent judge mismatch.

Q03 (correctness 0.428): the answer gives AED 75 and details exemptions; the short reference says AED 75 subject to exemptions. Its faithfulness is 1.000. The extra supported detail may be lowering correctness relative to the short reference.

Q02 (correctness 0.512): the answer adds a comprehensive-policy requirement and cites a second document. Review whether that extra claim belongs to the law asked about; faithfulness over all five selected chunks cannot by itself verify law-specific attribution.

Q07 (relevancy 0.708): the answer adds a tag requirement beyond the toll-fee trigger asked for. The lower focus score is plausible.

Q08 and Q09: /chat returned reranker model_failure (HTTP 502) on first attempt and retry. No answer or Ragas metrics were produced.

Three negative/status cases behaved as expected: N01 insufficient_evidence, N02 out_of_scope, and N03 needs_clarification. Follow-up continuity and thread isolation passed.

Search modes: separate retrieval results

Each mode received the same ten original questions through /search. These are retrieval outcomes, not Ragas answer scores.

Mode	Document hit@5	Exact chunk hit@5	Document at rank 1	Mean request time
Lexical	10/10	10/10	9/10	1.47 s
Semantic	10/10	10/10	10/10	1.77 s
Hybrid	10/10	10/10	10/10	2.00 s

Lexical search

SQLite FTS5/BM25 matched words in chunk text. It found every expected document and exact chunk within five results. Q02 was rank 2; the other nine expected documents were rank 1. Mean request time was 1.47 s across these ten sequential calls.

Semantic search

The embedding model converted each question to a vector, and cosine similarity ranked matching chunks. Every expected document and exact chunk was rank 1. Mean request time was 1.77 s.

Hybrid search

Lexical and semantic branches ran together. Reciprocal rank fusion merged and deduplicated candidate chunks, then a separate LLM call reranked them. Every expected document and exact chunk was rank 1. Mean request time was 2.00 s.

Rank by question (D = expected document; C = exact expected chunk)

Case	Lexical D/C	Semantic D/C	Hybrid D/C
Q01	1/1	1/1	1/1
Q02	2/2	1/1	1/1
Q03	1/1	1/1	1/1
Q04	1/1	1/1	1/1
Q05	1/1	1/1	1/1
Q06	1/1	1/1	1/1
Q07	1/1	1/1	1/1
Q08	1/1	1/1	1/1
Q09	1/1	1/1	1/1
Q10	1/1	1/1	1/1

The observed time difference is descriptive, not a controlled latency benchmark. It includes network model calls and was measured once per question. The modes tied on hit@5; this run does not establish that hybrid is better than semantic on unseen questions.

Outcome and conclusions

What this run supports, and what should happen next

1. Retrieval found the known sources

All three modes achieved 10/10 expected-document and exact-chunk hit@5 on these curated questions. Semantic and hybrid placed all ten expected sources first; lexical placed nine first. The questions were written from the indexed corpus, so this is evidence for those cases rather than a general accuracy estimate.

2. End-to-end reliability needs work

Only 8/10 source questions received a first-attempt /chat answer. Q08 and Q09 failed at the reranker twice each, even though their sources were found in separate search calls. The priority engineering issue is to diagnose and make reranker output validation/recovery reliable, then repeat this evaluation.

3. The judge needs calibration

The eight scored answers had high average faithfulness, precision, recall, and relevancy, but mean answer correctness was 0.650. Q06 and Q10 show that a low judge value can coexist with a closely matching cited answer. Use the per-case answers and passages to review low values. The same gpt-4.1-mini deployment family answered and judged these cases.

4. Expand the evidence before setting thresholds

Add more unseen, human-reviewed legal questions with multi-claim references, difficult table rows, exceptions, and ambiguous requests. Repeat each run to measure model variability. Keep retrieval hit@k, answer quality, citation support, abstention behavior, and API failures as separate measures.

Method and limits

Run ID 20261111_205750; Ragas 0.4.3; gpt-4.1-mini evaluator; text-embedding-3-small; final top_k=5. Chat Ragas contexts were taken from the exact checkpointed evidence. Only answered first attempts enter the five Ragas means (n=8). The 13-case status set had 11/13 expected statuses; the three negative cases passed. Mean /chat request time was 5.14 s and median 6.11 s, excluding Ragas judging. These values are from one local run.

Evidence: evals/ragas/results/20261111_205750/results.json (raw responses and scores), report.md (aggregate table), review.md (manual review), and evals/ragas/cases.json (questions and references).